9.12 Random Subsets

WORKSHEET

The question

STEP III 2019, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How can you describe a subset of $\{1, \ldots, n\}$ by a sequence of n answers to yes/no questions?

STAGE 2 Disjoint sets

Ask yourself For $A_1 \cap A_2 = \varnothing$, what must be true of each element i? What is the probability of that, for one element? Why can you multiply over the elements?

STAGE 3 Subsets

Ask yourself For $A_1 \subseteq A_2$, which of the four possibilities for an element is forbidden?

Ask yourself For $A_1 \subseteq A_2 \subseteq A_3$, which of the 8 patterns of membership (in or out of each set) are allowed for an element?

What's really going on

Counting without probability

Multiplying by the number of choices turns each probability into a count. The number of ordered pairs (A_1, A_2) with $A_1 \subseteq A_2$ is $4^n \times (\frac{3}{4})^n = 3^n$. You can also count these pairs directly: choose the size k of A_2, then A_2 itself in $\binom{n}{k}$ ways, then A_1 as any of its 2^k subsets. So

$$\sum_{k=0}^{n} \binom{n}{k} 2^k = 3^n,$$

which is the binomial theorem for $(1+2)^n$. Counting the same thing in two ways, once by elements and once by sizes, proves an identity: this is the method of *double counting*, which you met with the Platonic solids in Section 7.9. Similarly, the number of chains $A_1 \subseteq \cdots \subseteq A_m$ is $(m+1)^n$.

The lattice of subsets

The subsets of $\{1, \ldots, n\}$, ordered by inclusion, form a structure called the *Boolean lattice*, which can be pictured as an n-dimensional cube: each subset is a corner, labelled by its sequence of yes/no answers, and moving along an edge adds or removes one element. Chains are paths going upwards. Deep results describe how large families of subsets can be, given conditions on them. *Sperner's theorem* (1928) says that the largest family in which no set contains another has $\binom{n}{\lfloor n/2 \rfloor}$ sets; the *Erdős–Ko–Rado theorem* (1961) says how many k-element sets can be chosen so that every two overlap. Both are standard in a first course on combinatorics, and the coin-tossing view of subsets gives some of the slickest proofs.

Boole's algebra

George Boole's *The Laws of Thought* (1854) treated logical statements as sets and "and", "or" and "not" as intersection, union and complement. Encoding a set as a string of yes/no answers is exactly how a computer represents it: a subset of $\{1, \ldots, 64\}$ is a single 64-bit number, and intersection is the "bitwise AND" instruction. The method

of this question, one element (one bit) at a time, is how such operations are analysed.

Try this next

1 **Covering.** Find $\mathrm{P}(A_1 \cup A_2 = S)$ and $\mathrm{P}(A_1 \cup A_2 \cup \cdots \cup A_m = S)$.

2 **Same size.** Show that $\mathrm{P}(|A_1| = |A_2|) = \binom{2n}{n}/4^n$. (*Hint:* $|A_1| = |A_2|$ exactly when $|A_1| + |S \setminus A_2| = n$.) Show that this is about $1/\sqrt{\pi n}$ for large n.

3 **Partitions into three.** Show that the number of ordered pairs of disjoint subsets (A, B) of S is 3^n, and find the number of unordered pairs $\{A, B\}$ of disjoint subsets.

4 **Sperner for small n.** For $n = 3$ and $n = 4$, find the largest family of subsets of $\{1, \ldots, n\}$ in which no set contains another.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on subsets and partially ordered sets.
- **Béla Bollobás, *Combinatorics: Set Systems, Hypergraphs, Families of Vectors and Combinatorial Probability*** (Cambridge University Press). A short, elegant book on families of subsets, from Sperner's theorem onwards.
- **Peter Cameron, *Combinatorics: Topics, Techniques, Algorithms*** (Cambridge University Press). A first course with a strong emphasis on counting in two ways.

The full guided solution is in Section 9.12:
https://beyondstep.keithmcnulty.org/ch09/random-subsets.html